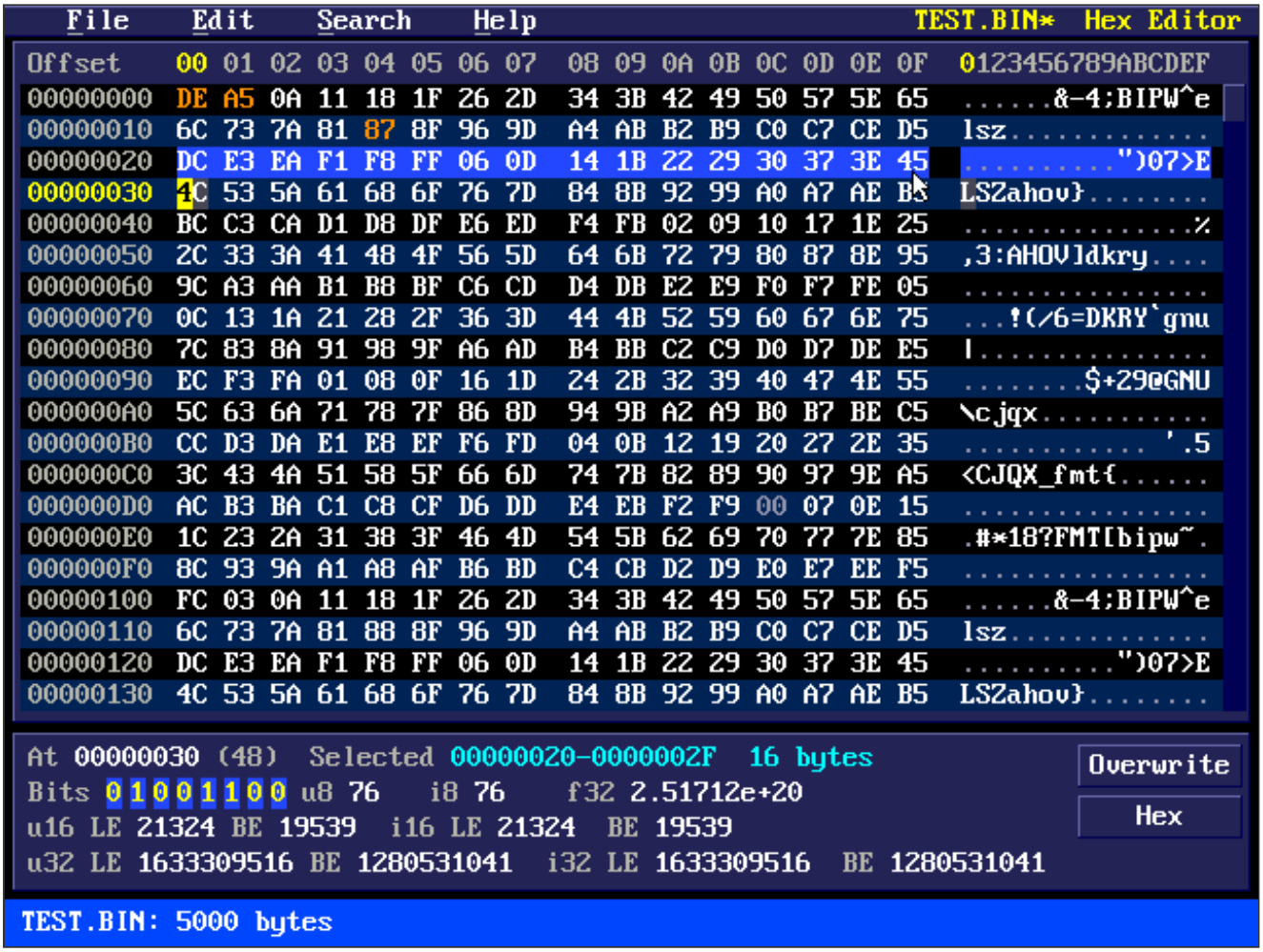
Pico 2 mini OS Hex Editor

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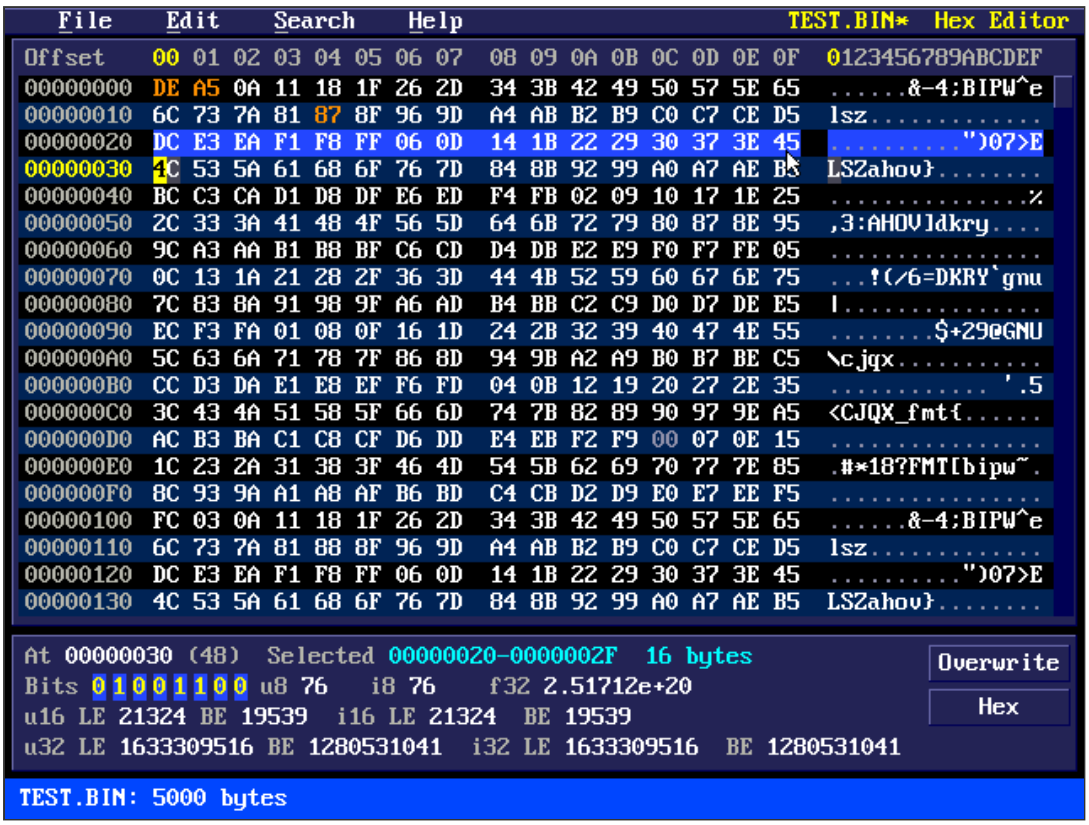
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A hex editor for any file on the drive: its bytes in hex and as text, overwritten, inserted, deleted, cut, copied and pasted with the keys or the mouse, with Find, Go to, undo and a panel that reads the bytes at the cursor as bits and numbers. Files of any size up to 4 GB open at once: the file stays on the drive and is read as it is shown.

The editor



`hex` (or `hexedit`) opens the editor on an empty, untitled file; `hex GAME.BIN` opens that file (or starts a new one by that name). It works with the keyboard alone; a USB mouse helps.

- **The menu bar** at the top: File, Edit, Search and Help, as in the tracker. Click a title, or press F10 (or F9), or Alt and the title's first letter. In a menu, Up and Down choose, Left and Right go to the next menu, Enter or the underlined letter runs an item, Esc closes it. Items that can't be used now are grey. The file's name is on the right, with `*` when it has changes not saved.
- **The bytes**, 16 a row: the offset of the row (hex), the bytes in hex (a wider gap after the eighth) and the same bytes as text (`.` for those that aren't letters, digits or signs). Zero bytes are dim; bytes changed since the file was opened or saved are **orange**. The cursor is yellow on the digit or character being typed, and grey in the other column; its row's offset and its column's number are yellow too. The selection is blue. Rows go one past the end, so that bytes can be added there.
- **The scroll bar** on the right: drag the thumb, or click above or below it for a screen up or down. The wheel scrolls three rows.
- **The panel under the bytes:** where the cursor is (hex, and decimal), the selection (its first and last byte and its length) or, with none, the clipboard's length; the byte at the cursor as **bits** (click one to flip it) and as an unsigned and a signed 8-bit number; the four bytes from the cursor as a 32-bit float (little-endian); and the bytes from the cursor as 16- and 32-bit numbers, unsigned and signed, little-endian (LE, the low byte first, as the Pico, a PC and most files keep them) and big-endian (BE). A `-` means the

file ends first. **Overwrite/Insert** and **Hex/Text** on the right switch the typing mode and the column.
- **The status line** at the bottom: the size and the modes, or a message for a few seconds.

Typing

In the hex column, the hex digits 0-9 and A-F type a byte: the first digit is its top half, the second its bottom half, then the cursor moves on. In **overwrite** mode (the start) they change the byte at the cursor; in **insert** mode (Ins) the first digit puts a new byte in (its bottom half 0 until the second digit). A click on a byte's second digit changes just that half. At the end of the file typing adds bytes in either mode.

Tab (or a click there) moves to the text column, where letters, digits and signs type themselves as bytes, overwriting or inserting the same way.

Typing over a selection deletes it first. Del deletes the byte at the cursor, or the selection; Backspace deletes the byte before the cursor in insert mode, and only moves back in overwrite mode.

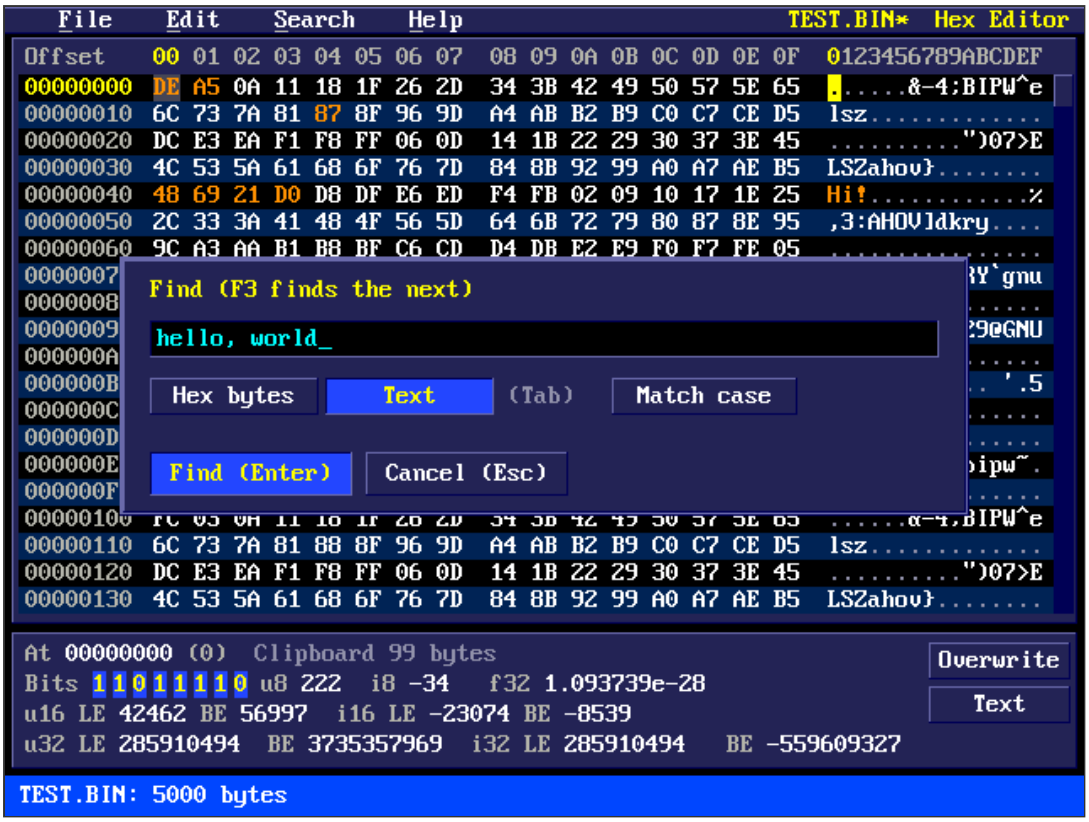
Selecting, cutting and pasting

Shift with the arrows (or Home, End, PgUp, PgDn) selects from where the cursor was; dragging with the mouse selects from the byte pressed on to the one under the pointer (past the top or bottom the rows scroll); Ctrl-A selects the whole file. Shift and a click select up to that byte.

Cut (Ctrl-X) and Copy (Ctrl-C) put the selection on the clipboard; Paste (Ctrl-V) puts it in place of the selection, or at the cursor: over the bytes there in overwrite mode, in front of them in insert mode. A cut or paste of any size is quick, as nothing is copied (see *How it works*). The clipboard stays while the editor runs, from one file to another; Save, Open and New keep it if it is up to 32 KB, and empty it if it is bigger.

The right button on the bytes opens the Edit menu there.

Find and Go to



Find (Ctrl-F, in the Search menu) looks for **hex bytes** (pairs of digits, spaces between them or not: `DE AD BE EF`) or **text** (Tab switches; *Match case* makes upper and lower case differ, which they don't at first). It looks from just after the cursor to the end, then from the start; what it finds is selected. With a short selection, Find starts with its bytes in it. **Find next** (F3) looks again. A search through a big file shows how far it has got; Esc stops it.

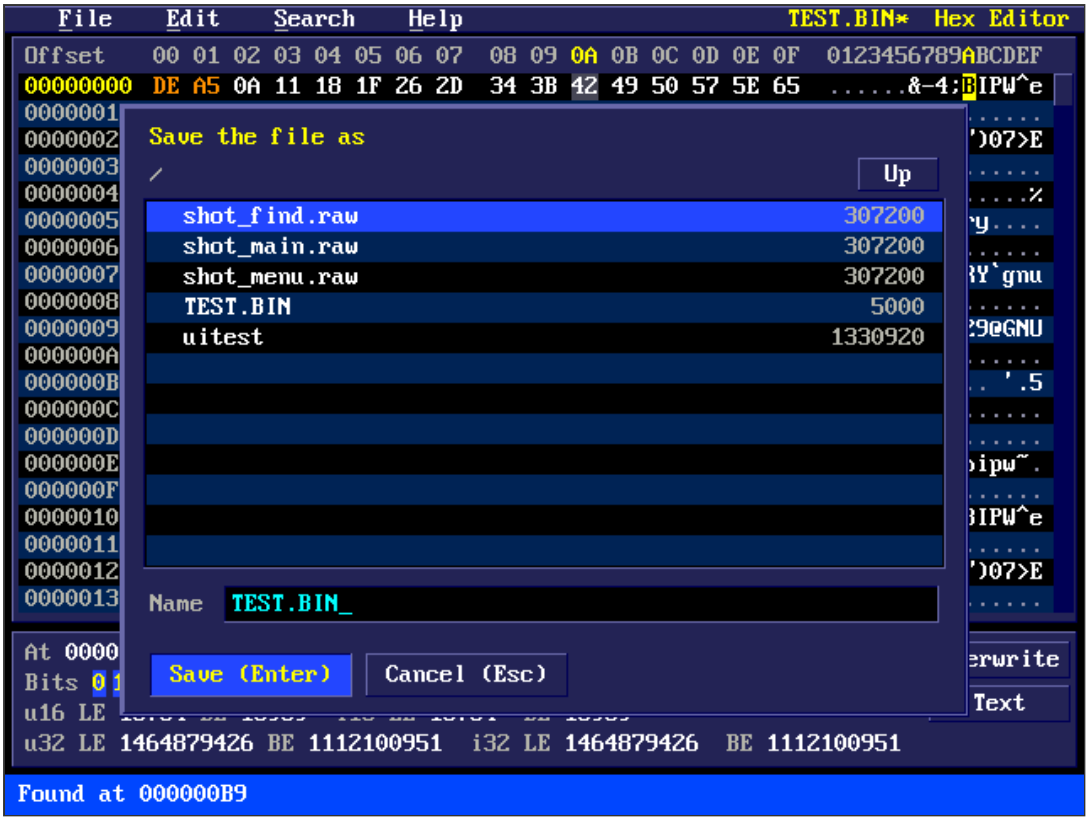
Go to offset (Ctrl-G) moves the cursor: an offset in hex (`1F4`, `0x1F4`, `$1F4` or `&H1F4`) or, after Tab, in decimal; `+` or `-` in front counts from the cursor. Home and End go to the row's start and end, and pressed again to the file's start and end.

Undo

Ctrl-Z undoes and Ctrl-Y redoes up to 64 steps: typing in one place is one step; a delete, cut, paste or bit flip is one each. Saving forgets them (the file they come from is replaced).

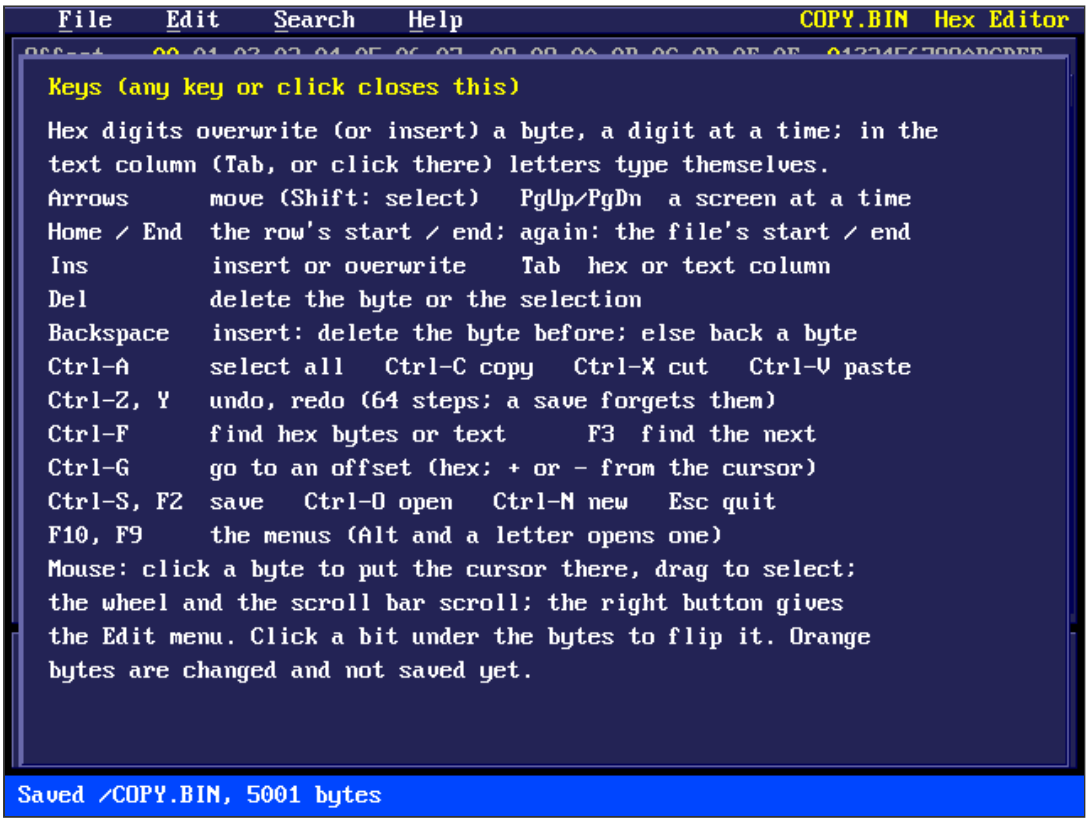
Files

Menu item	Key	Does
New	Ctrl-N	an empty, untitled file
Open...	Ctrl-O	the file box: any file
Save	Ctrl-S, F2	writes the file (asks for a name if it has none)
Save as...		under another name (asks before replacing a file)
Quit	Esc	leaves



New, Open and Quit ask first if there are changes not saved. The file box lists the folders and files with their sizes; Up (or Backspace with no name typed) goes to the folder above.

Keys



Help, Keys (F1) shows them in the editor too.

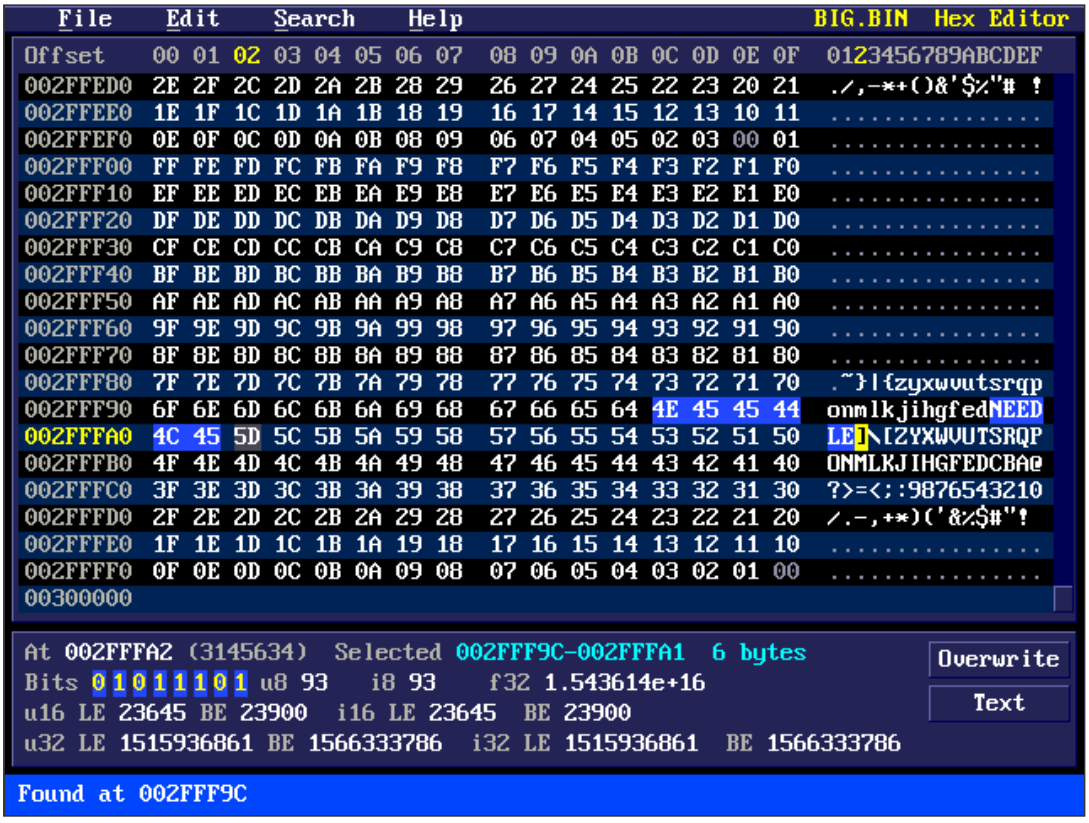
Key	Does
Arrows	move a byte or a row (Shift: select)
PgUp, PgDn	a screen up or down
Home, End	the row's start or end; again: the file's start or end
0-9, A-F	type in the hex column
Tab	the hex or the text column
Ins	insert or overwrite
Del, Backspace	delete (see *Typing*)
Ctrl-A, C, X, V	select all, copy, cut, paste
Ctrl-Z, Ctrl-Y	undo, redo
Ctrl-F, F3, Ctrl-G	find, find next, go to
Ctrl-S or F2, Ctrl-O, Ctrl-N	save, open, new
F10, F9, Alt-letter	the menus
F1	the keys
Esc	quit

How it works

The file stays on the drive: the editor reads it 512 bytes at a time through a cache of eight such pages, and reads more at once when it saves or searches. The changes are kept apart in a *piece table*: the file as it is now is a list of pieces, each a run of bytes either from the file or from the bytes typed (kept in a buffer that only grows). Typing adds bytes to that buffer; a delete, cut or paste only changes the list, which is why they are quick for any size, and the clipboard is pieces too. An undo step keeps the pieces it replaced.

All of that is in SRAM, taken when the editor starts and given back when it quits (the OS keeps no memory for it otherwise). It has room for about 16,000 pieces and up to 192 KB typed between saves, as SRAM allows; when it runs short the oldest undo steps make way, and if that isn't enough the editor says so: save, and it goes on from there. Only the picture drawn off the screen is in PSRAM.

Save writes the whole file anew, to `HEXEDIT.TMP` in the same folder, then deletes the old file and renames the new one to its name, so a save that fails (a full drive, or Esc during a long one) leaves the file as it was. Saving a big file takes as long as copying it.



Notes

- Bytes the drive can't read show as `00`, with a message; Save then stops rather than write them.
- Files of 4 GB and more (possible on exFAT) don't open.
- The PC test (`tools/hextest/uitest.sh [folder]`) runs the editor on a PC with thousands of random edits checked against a plain copy, undo and redo all the way back and forth, saves, Find, Go to, the menus and the mouse, and writes the screenshots in this manual.